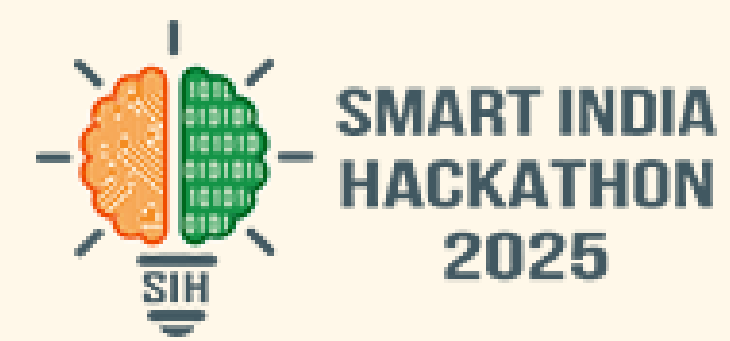




SMART INDIA HACKATHON 2025



TITLE PAGE

Problem Statement Title -

**STUDENT INNOVATION: SWADESHI FOR ATMANIRBHAR
BHARAT - ROBOTICS AND DRONES (SOFTWARE)**

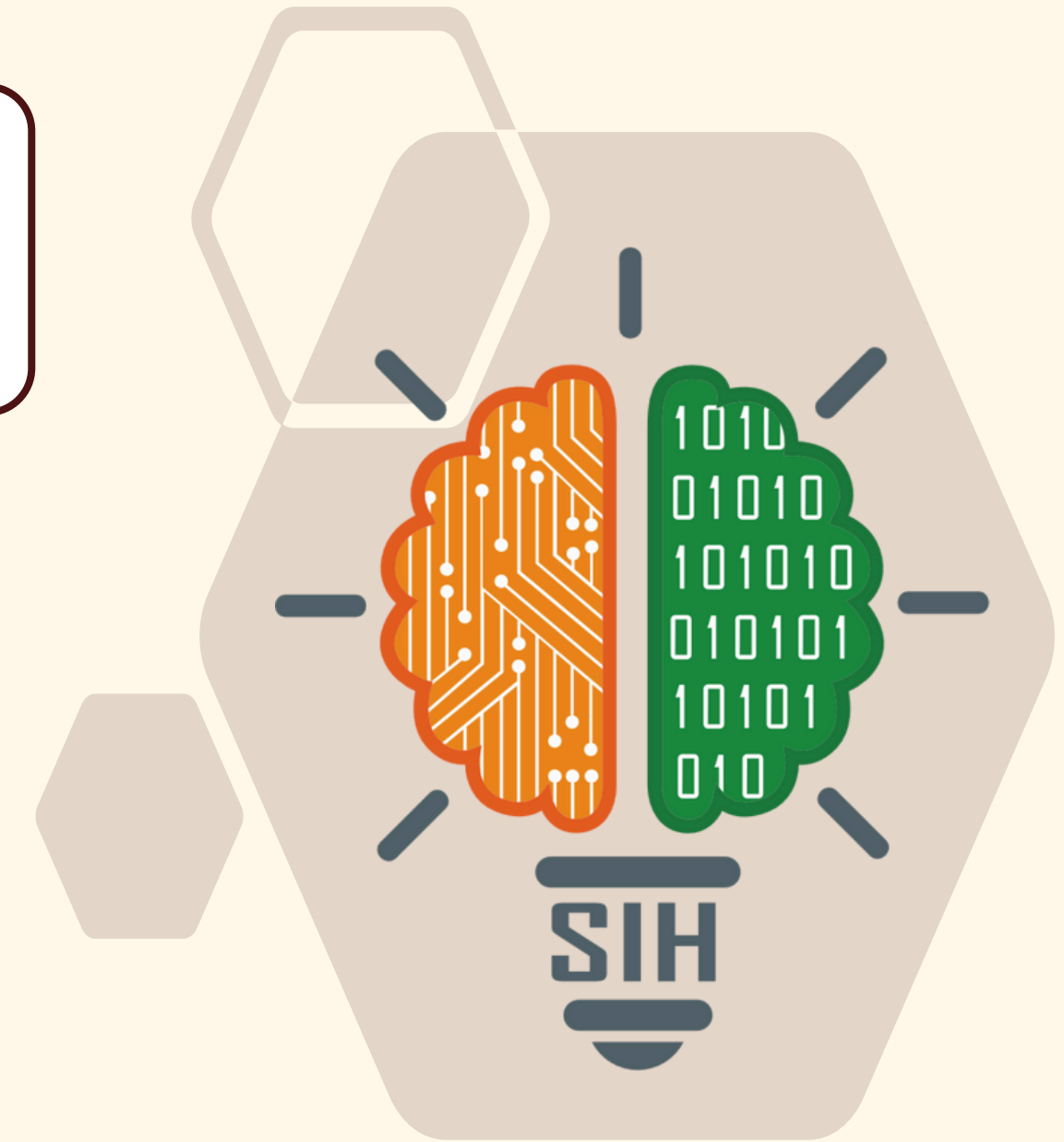
Problem Statement ID - SIH25134

Theme - Robotics and Drones

PS Category- Software

Team ID - 83470

Team Name - Team M3OW (MACHINES, MATHEMATICS, AND MODELS OF WONDER)





SANJEEVANI UAV - UAV CASUALTY TRIAGE AND RESPONSE SYSTEM

PROPOSED SOLUTION

We propose an AI-automated pipeline for search, triage, and rescue in Mass Casualty Incidents (MCIs) such as natural calamities, bombings, and battlefields by harnessing Unmanned Aerial Vehicles (UAVs). Using multimodal sensing and zero-shot AI processing, UAVs can autonomously detect, localize, and assess casualties. Through automated triage, the system estimates injury severity and generates real-time casualty maps to assist medics, improving decision-making, survival outcomes, and the overall efficiency of emergency response operations.

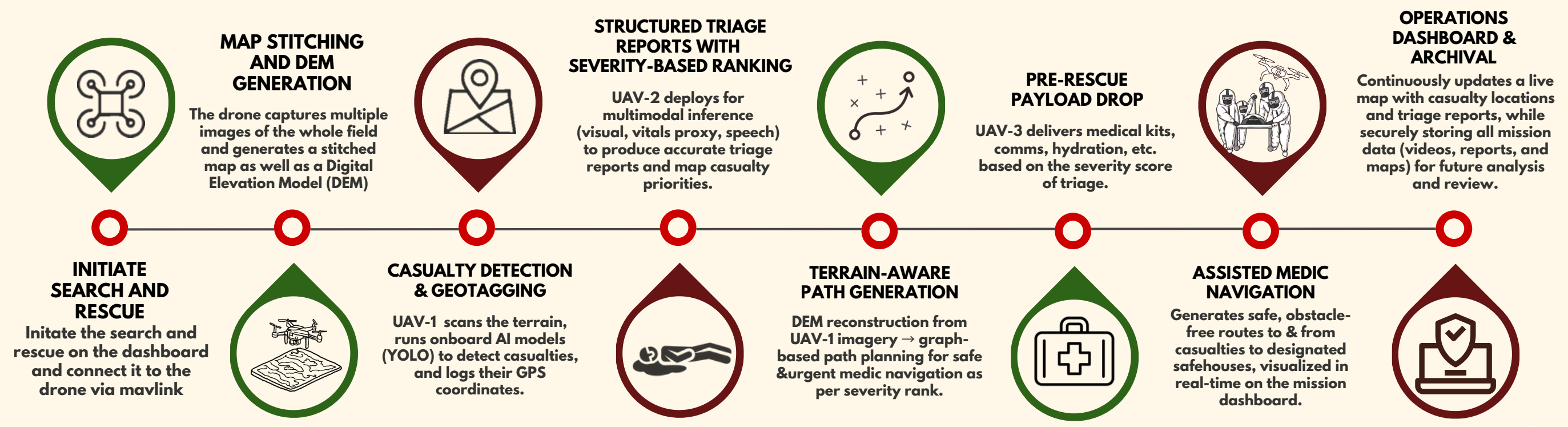
FEATURES

- 01 AUTOMATED LOCALIZATION & TRIAGING OF CASUALTIES
- 02 RISK & PRIORITY ASSESSMENT OF CASUALTIES BASED ON THE SEVERITY OF THE INJURIES
- 03 AUTONOMOUS DRONE FLIGHT AND MAPPING, WITH PAYLOAD DELIVERY TO CASUALTIES
- 04 RESCUE PATH PLANNING FOR RAPID MEDIC ACCESS AND CASUALTY EVACUATION.



USP - WHY WE ARE BETTER ?

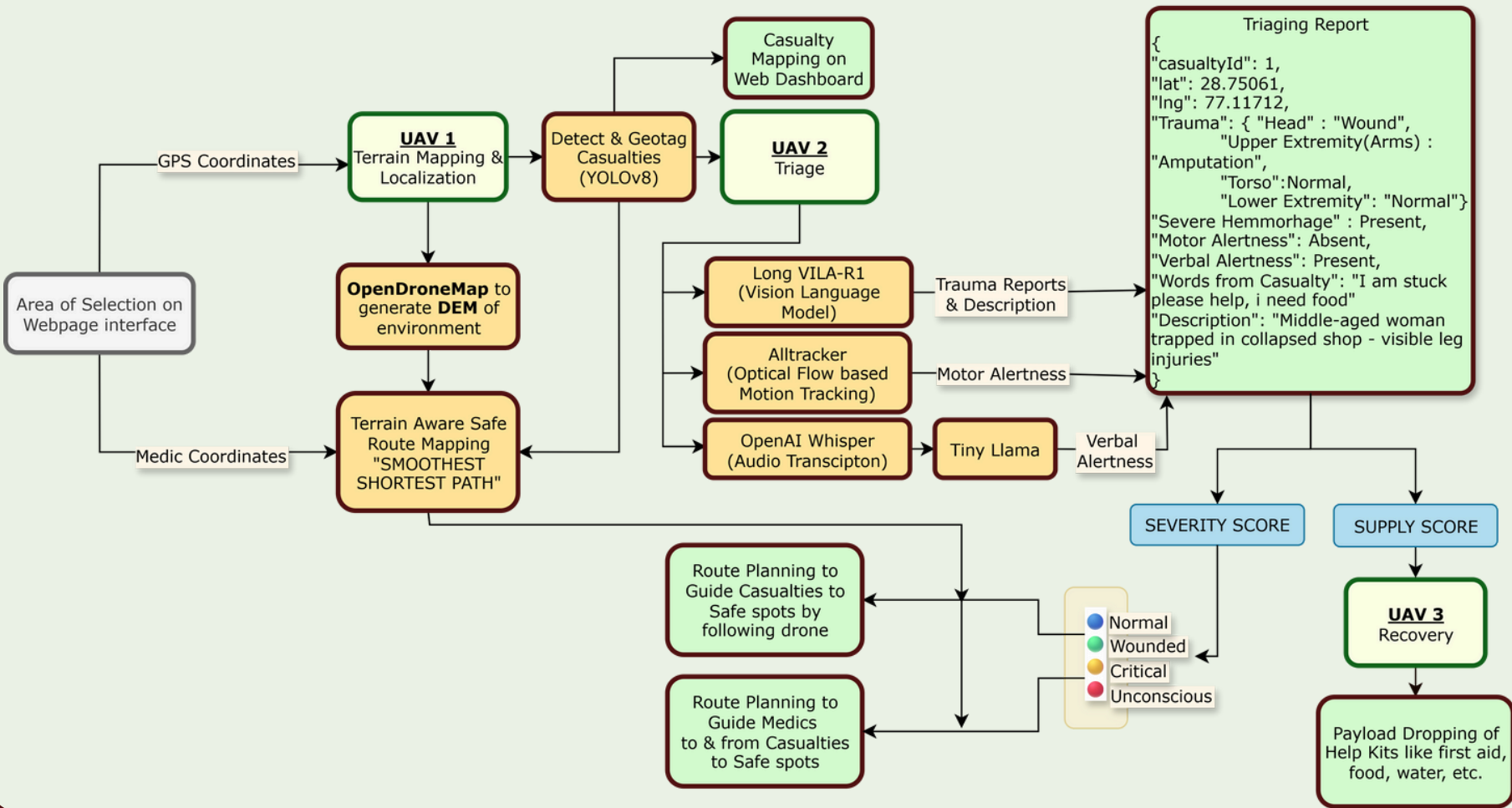
- MULTI LINGUAL
- NO GEOGRAPHICAL BIAS
- SCALABLE TO CURRENT UAV ARCHITECTURES
- TERRAIN AWARE PATH MAPPING FOR MEDICS
- RISK AND PRIORITY BASED RESPONSE
- POST OPERATION ANALYSIS AND TRAINING



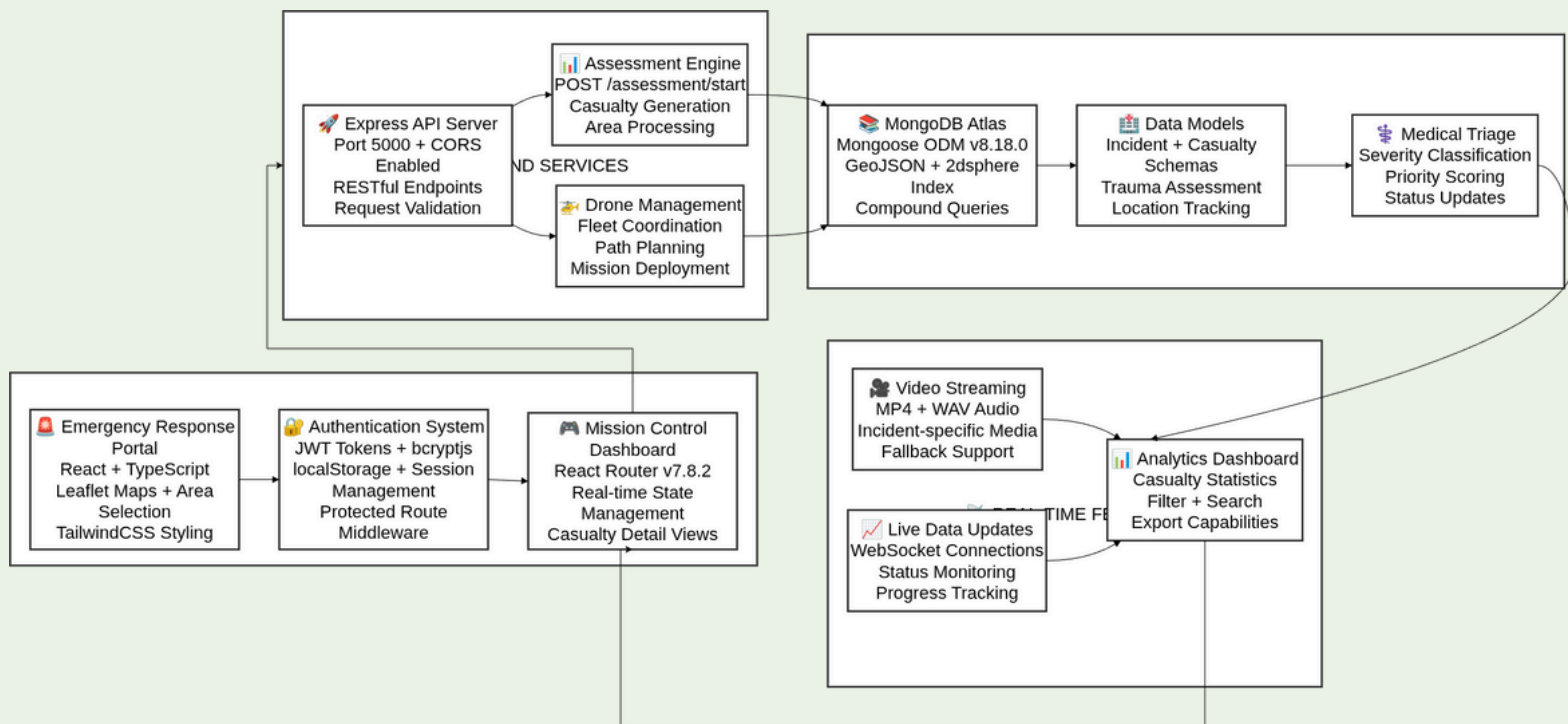
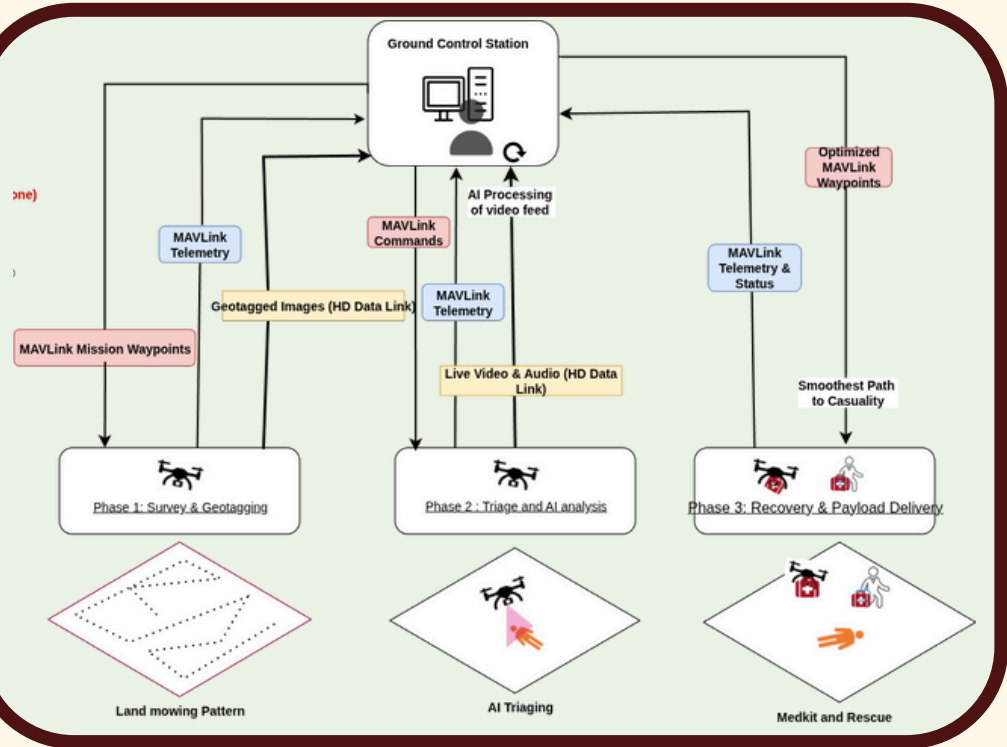


TECHNICAL APPROACH

TECH STACK



- **OpenDroneMap (ODM):** UAV 1 processes aerial imagery to generate DEMs from Structure from Motion, aiding, localization, and shortest-path planning.
- **Vision-Language Model (VILA-R1):** UAV 2 analyzes casualty images, producing a detailed trauma report and description.
- **AllTracker:** Uses optical flow for Dense Point Tracking to monitor movement, providing motor alertness classification.
- **Whisper + TinyLlama Pipeline:** Whisper transcribes audio, TinyLlama processes it, generating a comprehensive triaging report to support informed rescue operations .



FRONTEND	
TypeScript	TS
Tailwind CSS	
PostCSS	
Vite	
shadcn/ui	
MAPPING	
Leaflet	Leaflet
React Leaflet	
Mapbox	mapbox
BACKEND	
Node.js	node
Express.js	express
MongoDB	mongoDB
Mongoose (WebServer)	
AI	
PyTorch	
Python	
Llama	LLAMA
Whisper	
LONG-RL	LONG-RL
DRONE CONTROLLER	
MAVLink	MAVLink
ArduPilot	ARDUPILOT



FEASIBILITY AND VIABILITY

CHALLENGE

Drones are expensive investments, How is the operation feasible in economic constraints ?

- Drones are becoming more affordable and accessible
- The ROI is worth the investment - some minor amount may be very worth saving lives
- We are currently working on production and fabrication of 3D printed drones of ourselves, and have successfully tested the initial prototypes.(refer last slide)

STRATEGY

CHALLENGE

Our Problem Statement is a very critical, is it ethical to rely on AI models for it ?

- A human operator always retains final decision-making authority.
- Livestream of Drone triage and reports are sent to the Ground Control Station at all times for manual confirmation
- The final control of the drone is always in hand of a Drone Operator/Pilot in case autonomy fails

STRATEGY

CHALLENGE

Considering hallucinations and small errors that may lead to wrong triage predictions, How do we plan on dealing with them ?

Regarding Hallucinations, mostly MLLMs are susceptible to this problem, we are actively researching into mitigation of the issue by our own experimentations and implementing of A* conference research papers using techniques like Logit Lens, Answer to Image attention, etc.

STRATEGY

CHALLENGE

The Drones and portal maybe hacked leaking essential data. How do we deal with it ?

- All the data transfer takes over a private network, set up at GCS.
- All the communication of reports between the Drone to GCS are AES256 encrypted.

STRATEGY

CHALLENGE

How does it compare to the current Triage Methods used in Search and Rescue operations?

- Our system weaves into the traditional methods by pre-mapping casualties and tagging them with AI-derived Severity + Resource scores.
- We also provide terrain maps, and safely traversable routes for the rescue teams mapped according to the casualty severity

STRATEGY

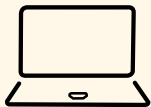
CHALLENGE

What is the current process of autonomy ? How do you plan on dealing with obstacles and debris ?

We are actively working and testing into the different aspects of Vision Language Action by the means of World Foundation Models(e.g NVIDIA Cosmos) and Vision World Models(VJEPa2) and its implementations possible in the field of drone autonomy, and plan to use that in the future

STRATEGY

FEASIBILITY ANALYSIS



TECHNICAL VIABILITY

- The system is technically viable by leveraging existing, developed technologies, like Mavlink, Ardupilot, OpenDroneMap, etc.
- The multi-modal approach, allows for a more comprehensive and accurate assessment than traditional, single-source methods.



OPERATIONAL VIABILITY

- The system's operational viability is strong as it directly enhances the speed and effectiveness of emergency response.
- The ability to autonomously deliver aid and guide first responders to exact locations streamlines the entire rescue workflow, ultimately saving valuable time and lives.



MARKET VIABILITY

- The market for this system is highly receptive, as government and public safety agencies are actively seeking technological solutions to improve disaster response.
- Its value proposition of increasing efficiency and safety makes it a compelling investment, justifying the initial cost of deployment.

IMPACT AND BENEFITS

KEY PARTNERS

1. National Level:

- National Disaster Management Authority (NDMA)
- National Disaster Response Force (NDRF).

2. Defense & Paramilitary:

- Indian Army Medical Corps
- Border Security Force (BSF).

3. State Level:

- State Disaster Response Forces (SDRFs)
- Police & Fire Departments.

CRITICAL NATIONAL NEED

The 2025 Uttarakhand flash floods, gave a total death count and missing counts comes to 70+ adults, Even after the heroic actions of NDRF teams, some people deceased due to lack of quick help and assesment. We need smart and fast response systems for saving more people

PROBLEMS ??

- High Initial Investment
- Projectiles thrown at drone
- Scope of Human Operator Errors
- Need to Define Legal and Ethical Boundaries
- Is Mavlink secure ?

DIVERSE APPLICATIONS

National Disasters:

- Earthquakes
- floods
- wildfires
- landslides.

Industrial & Urban Incidents:

- Building collapses
- factory explosions
- major transport accidents.

Hazardous Environments

- Open-pit mining rockfalls
- Mine collapses
- Chemical spills

Defense & Security:

- Battlefield triage
- post-blast assessment
- border security incidents.

Remote Search & Rescue (SAR):

- Wilderness
- Maritime
- High-altitude mountaineering.

Humanitarian Crises:

- Rapid needs assessment in conflict zones and refugee camps.

FUTURE SCOPE

- **Cheaper "Make in India" Drones:** Our team is actively working with Designers for the synthesis of 3D printed drones.
- **VLA Based Autonomy and Projectile Avoidance:** Our next phases of work involve working on Drone Autonomy based on Vision Language Action, instead of the primitive mapping methods, also we are trying to look into methods to make the drone avoid projectiles like stones pelted towards it using VLA
- **UxV Systems for indoor triage:** We are planning to use a combination of UGV and UAV systems, to perform triage in spaces, where a drone is unable to reach with convenience like indoor debris.
- **Data Security:** The later phases of the project will involve working on Drone Security and Data Transfer Security, instead of the current AES encryption.
- **Drone Swarm Capabilities:** The eventual phases will demand increase in the scale of the project, hence instead of a single uav performing the operation, we may use swarms of drones to cover larger areas more efficiently

TIME OF TRIAGE VS GOLDEN HOUR

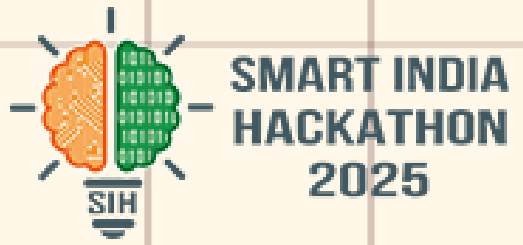
- The "Golden Hour" is a concept first described by surgeon R Adams Cowley, postulating that a trauma victim's best chance for survival occurs when they receive definitive surgical care within 60 minutes of injury. Traditional Triage and Rescue operations are way longer and miss this golden hour window due to constraints in localisation, assesment and mapping.
- We aim at rescuing each casualty under the golden hour, so we can minimise the deaths in any MCI(Mass Casualty Incident)

IMPACT AND BENEFITS

1. AI-Driven Standardization of Triage Decisions
2. Reduced Medic Burnout & Cognitive Load
3. Early Victim Stabilization via Payload Delivery
4. Data for Post-Disaster Forensics & Training
5. Access to Hazardous & Inaccessible Zones
6. Psychological Support & Assurance for Survivors
7. Scalable and Modular Deployment Architecture



RESEARCH AND REFERENCES



DEMOS AND VIDEOS

- Video Playlist : <https://tinyurl.com/ytm3ow>
- Our Website : <https://sanjeevniuav.hacker101.xyz>
- 3D printed drone video : <https://tinyurl.com/icarusuav>
- Post Analysis Dashboard of Test run:
<https://sanjeevniuav.hacker101.xyz/incident>

DOCUMENTS

- Technical Report Doc : <https://tinyurl.com/sanjeevniuavdoc>
- Github : <https://github.com/Team-M3OW/UAV-Triage-System>

REFERENCES

- Scaling RL to Long Videos(Chen et Al.), NVIDIA
- AllTracker: Efficient Dense Point Tracking at High Resolution(Harley et Al.), ICCV 2025
- LongVILA: Scaling Long-Context Visual Language Models for Long Videos(Chen et Al.), ICLR 2025
- Robust Speech Recognition via Large-Scale Weak Supervision(Radford et Al.), OpenAI
- OpenDroneMap